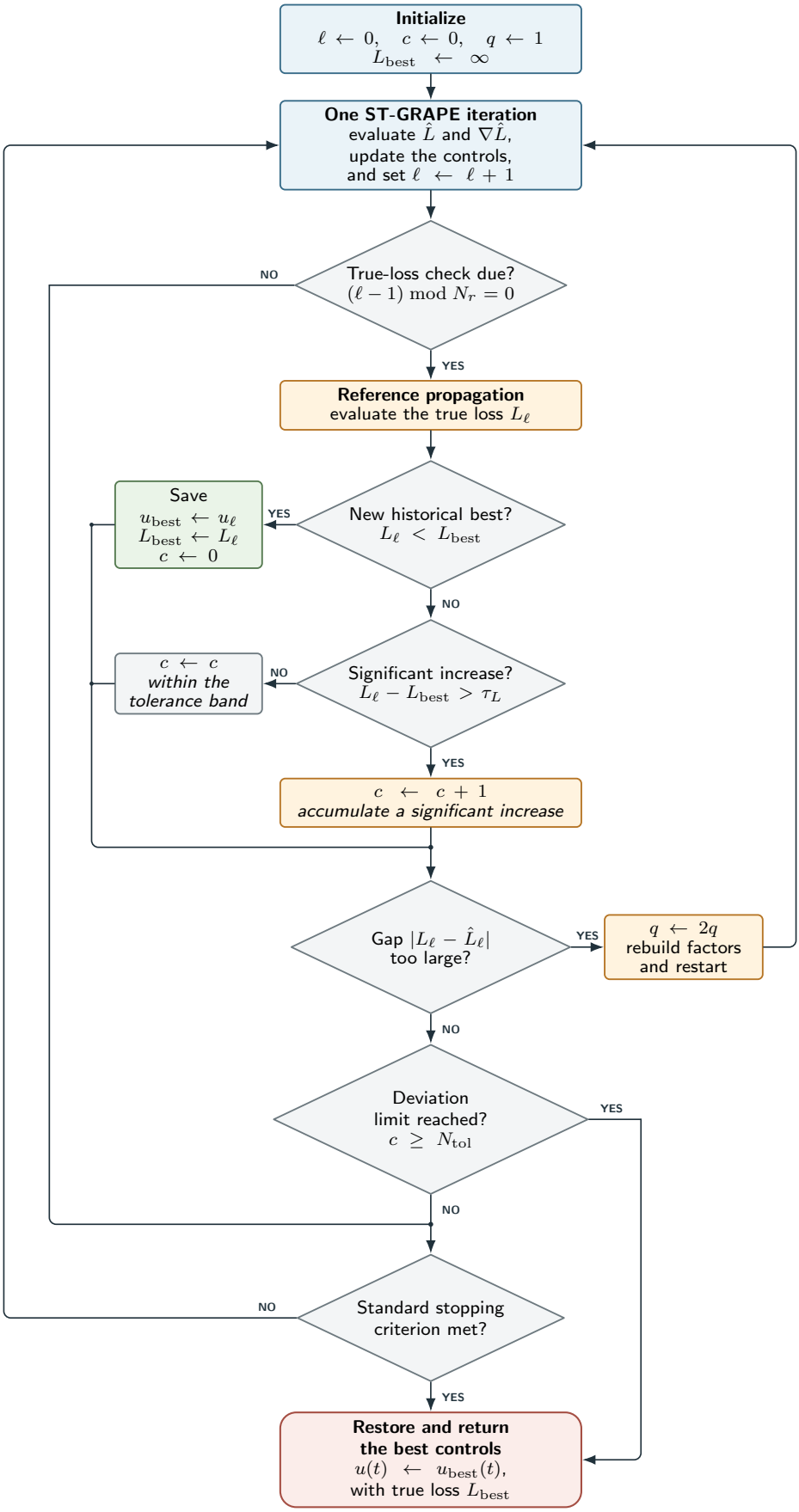

Initialize
$\ell \leftarrow 0, \quad c \leftarrow 0, \quad q \leftarrow 1$
$L_{\text{best}} \leftarrow \infty$
One ST-GRAPE iteration
evaluate $\hat{L}$ and $\nabla \hat{L}$,
update the controls,
and set $\ell \leftarrow \ell + 1$
True-loss check due?
$(\ell - 1) \bmod N_r = 0$
NO
YES
Reference propagation
evaluate the true loss L_ℓ
New historical best?
$L_\ell < L_{\text{best}}$
YES
Save
$u_{\text{best}} \leftarrow u_\ell$
$L_{\text{best}} \leftarrow L_\ell$
$c \leftarrow 0$
NO
Significant increase?
$L_\ell - L_{\text{best}} > \tau_L$
NO
$c \leftarrow c$
within the
tolerance band
YES
$c \leftarrow c + 1$
accumulate a significant increase
Gap $|L_\ell - \hat{L}_\ell|$
too large?
YES
$q \leftarrow 2q$
rebuild factors
and restart
NO
Deviation
limit reached?
$c \geq N_{\text{tol}}$
YES
NO
Standard stopping
criterion met?
NO
YES
Restore and return
the best controls
$u(t) \leftarrow u_{\text{best}}(t)$,
with true loss L_{best}